

Yes. If “Buck’s bound” means the printed Buck–Simon estimate

$$E_N(\gamma) \leq 11 L^{4/3},$$

then a sharper use of the same tube-packing argument gives

$$E_N(\gamma) \leq 4.724 L^{4/3}.$$

Combining this with Buck–Simon’s crossing-energy inequality gives the stronger universal ropelength lower bound

$$\text{Rop}(K) \geq (4\pi/4.724)^{3/4} c(K)^{3/4}$$

and numerically

$$\text{Rop}(K) \geq 2.08293 c(K)^{3/4}.$$

This improves the Buck–Simon printed constant

$$(4\pi/11)^{3/4} \approx 1.10500.$$

The exponent is the same; the improvement is in the coefficient.

1. Setup

Normalize the knot or link γ to have thickness 1. Then its ropelength is its length:

$$L = \text{length}(\gamma) = \text{Rop}(\gamma).$$

Let $E_N(\gamma)$ denote Buck–Simon’s normal energy. They prove

$$E_N(\gamma) \geq E_S(\gamma) \geq 4\pi \text{acn}(\gamma) \geq 4\pi c(K),$$

where $c(K)$ is the crossing number of the knot or link type. They also define ropelength as arclength divided by tube radius, and their printed 3/4-power estimate is $11E_L(K)^{4/3} \geq E_N(K)$. (Academia: https://www.academia.edu/103900587/Thickness_and_crossing_number_of_knots)

So the task is to replace

$$E_N \leq 11L^{4/3}$$

by a smaller constant.

2. Sharpening the Buck–Simon shell estimate

Fix a point $x \in \gamma$, and write the inner normal-energy integral as

$$I_x = \int_{\gamma} |T_x \times r_{xy}|^2 / |x-y|^2 dy,$$

where T_x is the unit tangent at x , and $r_{xy} = (y-x)/|y-x|$.

$$\text{Then } E_N(\gamma) = \int_{\gamma} I_x dx.$$

Split γ into a local and global part relative to x .

The local part consists of points on the same component whose arclength distance from x is at most π . Since the curve has thickness 1, Buck–Simon’s tangent-circle argument gives

$$|T_x \times r_{xy}|^2 / |x-y|^2 \leq 1/4.$$

The local part has length at most 2π , so

$$I_x^{\text{loc}} \leq 1/4(2\pi) = \pi/2.$$

For the global part, every point satisfies $|x-y| \geq 2$. For same-component points this is Buck–Simon’s Schur-theorem lemma; for points on other link components it follows from embedded unit tubes. Hence

$$I_x^{\text{glob}} \leq \int_{\{G_x\}} 1/|x-y|^2 dy.$$

Now define a measure μ_x on $[2, \infty)$ by

$$\mu_x([a, b]) = \text{length}\{y \in G_x : a \leq |x-y| \leq b\}.$$

The total mass of μ_x is at most $S = L - 2\pi$.

For $R \geq 2$, the unit tube around the part of G_x lying inside the ball of radius R centered at x lies inside the spherical annulus

$$1 \leq |p-x| \leq R+1.$$

Because the unit normal disks are disjoint, the tube volume is π times arclength. Buck-Simon use this same volume principle in their shell-packing estimate. (Academia: https://www.academia.edu/103900587/Thickness_and_crossing_number_of_knots) Therefore

$$\pi \mu_x([2, R]) \leq (4\pi/3)((R+1)^3 - 1),$$

$$\text{so } \mu_x([2, R]) \leq F(R), \text{ where } F(R) = 4/3((R+1)^3 - 1).$$

Because r^{-2} is decreasing, the largest possible value of $\int_{[2, \infty)} r^{-2} d\mu_x(r)$ under the constraints $\mu_x([2, R]) \leq F(R)$ and $\mu_x([2, \infty)) \leq S$ is obtained by packing as much length as allowed as close to radius 2 as possible. Equivalently,

$$I_x^{\text{glob}} \leq G(S),$$

$$\text{where } G(S) = \int_2^\infty \min(S, F(r)) \frac{2}{r^3} dr.$$

Now compute G . Since $F(2) = 4/3(27-1) = 104/3$, we have

$$G(S) = S/4 \text{ for } 0 \leq S \leq 104/3.$$

For $S \geq 104/3$, define $R \geq 2$ by

$$S = F(R) = 4/3((R+1)^3 - 1).$$

Then

$$G(S) = 26/3 + \int_2^R 4(r+1)^2/r^2 dr.$$

Since $4(r+1)^2/r^2 = 4 + 8/r + 4/r^2$, we get

$$G(S) = 26/3 + 4(R-2) + 8 \log(R/2) - 4(1/R - 1/2).$$

$$\text{Thus } G(S) = 8/3 + 4R + 8 \log(R/2) - 4/R.$$

Therefore, for every x ,

$$I_x \leq \pi/2 + G(L-2\pi).$$

3. Optimizing the coefficient

For $S = L - 2\pi \geq 104/3$, write

$$L(R) = 4/3((R+1)^3 - 1) + 2\pi,$$

$$\text{and } B(R) = \pi/2 + 8/3 + 4R + 8 \log(R/2) - 4/R.$$

We need to maximize $Q(R) = B(R)/L(R)^{1/3}$.

A direct derivative computation gives

$$Q'(R) = [4(R+1)^2/(3R^2 L(R)^{4/3})] (3L(R) - R^2 B(R)).$$

Let $H(R) = R^2 B(R) - 3L(R)$. Then $Q'(R)$ changes sign exactly when $H(R)=0$. One checks $H(2)<0$, and

$$H''(R) = \pi + 16 \log(R/2) + 16/3 > 0 \quad (R \geq 2).$$

Also $H(R) \rightarrow +\infty$. Thus H has a unique zero after 2, and Q has a unique maximum there. Numerically, interval evaluation gives

$$7.266757 < R_* < 7.266758,$$

$$\text{and } Q(R_*) < 4.724.$$

For the initial range $0 \leq S \leq 104/3$, the ratio $[\pi/2 + S/4]/(S + 2\pi)^{1/3}$ is smaller than this value. Therefore

$$I_x \leq 4.724 L^{1/3}$$

for every $x \in \gamma$. Integrating over x gives

$$E_N(\gamma) = \int_{\gamma} I_x \, dx \leq 4.724 L^{4/3}.$$

4. Crossing-number consequence

Now combine the lower and upper energy estimates:

$$4\pi c(K) \leq E_N(\gamma) \leq 4.724 L^{4/3}.$$

Hence $L^{4/3} \geq [4\pi/4.724] c(K)$. Therefore

$$L \geq (4\pi/4.724)^{3/4} c(K)^{3/4}.$$

Since L was the ropelength of an arbitrary thickness-1 representative, taking the infimum over all representatives gives

$$\text{Rop}(K) \geq (4\pi/4.724)^{3/4} c(K)^{3/4}$$

and $(4\pi/4.724)^{3/4} \approx 2.08293$. So a rigorously rounded version is

$$\text{Rop}(K) \geq 2.0829 c(K)^{3/4}.$$

For comparison, $(4\pi/11)^{3/4} \approx 1.1050$, so this is a substantial improvement over Buck-Simon's printed 11-constant bound, while retaining the same $c^{3/4}$ exponent.